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Synopsis

On

“TEXT TO SQL QUERY GENERATOR USING GEN - AI”

Submitted in partial fulfilment for the requirement of

Major Project [BCS685]

In

INFORMATION SCIENCE & ENGINEERING

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2026 – 2027



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CERTIFICATE

This is to certify that the major project work , entitled “**TEXT TO SQL QUERY GENERATOR USING GEN-AI**” is a Bonafide work carried out by **SWETHA , SAGAR SAWARN , YASHWARDHAN JHA** in partial fulfilment for the award of degree of Bachelor of Engineering in Information Science and Engineering during the academic year 2026-27.It is certified that all the corrections/suggestions indicated for Internal Assessment have been incorporated in the Report .The mini project has been approved as it satisfies the academic requirements.

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DECLARATION

We, **Swetha, Sagar Sawarn ,Yashwardhan Jha** hereby declare that the entire work titled **“TEXT TO SQL QUERY GENERATOR USING GEN-AI”** embodied in this major project report has been carried out by me during the 6th semester of BE(ISE) degree at BCET in Visvesvaraya Technological University, Belagavi under the esteemed guidance of **Mrs. Swathi Priya** Assistant Prof, Dept. of ISE, BCET. The work embodied in this dissertation work is original and it has not been submitted in part or full for any other degree in any University. Further we declare that the content of the dissertation has not been submitted previously by anybody for the award of any Degree or Diploma to any other University. We also declare that any Intellectual Property Rights generated out of this project carried out at BCET will be the property of BCET, Bangalore and we will be one of the authors of the same.

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ABSTRACT

With the aim of simplifying database interactions, this project introduces an innovative approach known as Semantic Synthesis, which transforms unstructured natural English language input into precise and contextually appropriate SQL queries. In response to the growing demand for user-friendly database access, cutting-edge Generative Artificial Intelligence (Generative AI) techniques are utilized to bridge the semantic gap between user-intended actions and the complexities of structured query language (SQL). The primary objective of this project is to empower individuals without extensive SQL expertise to interact with databases intuitively and extract knowledge from huge databases which help in making data-driven decisions, thereby enhancing usability and accessibility. To derive meaning and context from user-provided textual queries, the approach utilizes a Large Language Model (LLM) deeply integrated with transformers and a user-friendly User Interface (UI). The resulting system excels at generating SQL queries tailored to user-defined tasks and provides a detailed algorithmic breakdown to elucidate the intricacies of query development. This dual-dimensional output ensures transparency throughout the translation process and enhances comprehension of the artificial intelligence driven conversion from natural language to SQL. This study explores the effectiveness of Semantic Synthesis across various real-world domains, including E-commerce, healthcare, education, and human resources. The project attempts to prove the system's practical utility and potential impact in diverse professional settings by assessing its accuracy, efficiency, and user friendliness. This project aims to significantly contribute to the democratization of database interaction by adopting a comprehensive strategy that integrates technological innovation with practical usability, effectively adapting to the intricacies of different industries.

Keywords: Database, Structured Query Language (SQL), Generative AI, Artificial Intelligence (AI), Large Language Model (LLM)

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CHAPTER – 1

INTRODUCTION

In the world of modern data-driven environments, it is becoming more and more clear that a smooth and easily navigable database interface is required. This work initiates a novel endeavor by presenting the notion of Semantic Synthesis, an innovative method intended to remove any technical barriers such as technical proficiency or specialized knowledge in database interaction through the conversion of unstructured natural language text into accurate and contextually relevant SQL queries. The core component of this project is the use of sophisticated Generative Artificial Intelligence (AI) methods, which are carefully used to close the semantic gap between the complex semantics of structured query language (SQL) and the subtle representations of user meaning. The primary objective of this project is to enable people who might not be very proficient with SQL by giving them the means to interact with databases in an easy-to-use manner. In turn, this empowerment changes the user experience, making it more intuitive and user-friendly while also improving accessibility. The process behind this breakthrough is based on the use of transformers and LLM, which are deeply entwined with neural networks and deep learning models that are part of generative AI.

1.1 PROJECT OBJECTIVE

The task of translating natural language questions into Structured Query Language (SQL) statements, known as text to-SQL, has garnered significant attention within the fields of natural language processing and database management. This capability democratizes data access and analysis, enabling users to interact with databases without requiring in-depth knowledge of query languages. The development of AI-driven text-to-SQL systems has been critical in achieving this goal. [1]

Early approaches to text-to-SQL relied heavily on role based systems and semantic parsing techniques. These methods, while foundational, often struggled with the diversity and complexity inherent in natural language queries. The advent of deep learning and neural network models marked a significant shift, introducing sequence-to-sequence architectures that improved the translation of natural language to SQL [2].

The integration of large pre-trained language models (PLMs) and large language models has further advanced the field [3], enhancing the understanding of natural language semantics and the generation of accurate SQL queries. Recent surveys have highlighted the impact of PLMs on text-to SQL parsing, noting their ability to capture complex linguistic patterns and improve performance across benchmarks [4].

Despite these advancements, challenges remain, particularly in handling complex and cross-domain queries. The development of large-scale, human-labeled datasets, such as Spider, has been instrumental in evaluating and advancing text-to-SQL systems. These datasets provide diverse and complex queries that test the robustness and adaptability of current models. [1].

This survey aims to provide a comprehensive overview of the evolution of text-to-SQL systems, emphasizing the integration of artificial intelligence methodologies. We explore foundational concepts, current benchmarks, datasets, and models, offering insights into the advancements and challenges in the field. By examining the trajectory of text-to-SQL research, we aim to highlight the progress made and identify areas for future exploration.

1.1 MOTIVATION

SQL database interaction can be difficult for many people, especially those who are not in the database management or data science fields. The project’s goal is to lower barriers by offering a natural language interface, which will increase database accessibility for a larger user population. Driven by the goal of enabling non-technical users, the project aims to provide a system that enables people with little to no SQL experience to build queries with ease. It seeks to improve people’s interactions with the utilization of data by streamlining database interaction

1.2 NEED FOR TEXT TO SQL

Text-to-SQL systems provide a specialized solution for translating natural language queries into precise SQL statements, allowing users to interact directly with databases without requiring expertise in SQL syntax. While general purpose AI models like ChatGPT can assist in generating SQL queries, they often lack the domain-specific optimizations and accuracy that dedicated text-to-SQL systems are designed to offer.

These systems are tailored to manage complex database schemas and ensure the generation of syntactically correct and efficient SQL queries, significantly enhancing data retrieval and analysis processes. By focusing exclusively on the task of converting natural language to SQL, text-to-SQL systems.



Fig:1 Methodology for conducting the survey of Text-to-SQL systems.

deliver more reliable and contextually appropriate results. This makes them invaluable in scenarios where precise data manipulation is critical, such as in healthcare, finance, and business intelligence. Furthermore, the development of such systems incorporates advancements in natural language understanding, database schema modeling, and semantic parsing, contributing to their robustness and usability across diverse application domains.

FOUNDATIONS OF TEXT-TO-SQL

Text-to-SQL systems are designed to translate natural language queries into Structured Query Language (SQL) statements, enabling users to interact with databases without requiring expertise in SQL syntax. The foundational components of these systems as shown in Figure 2 include:

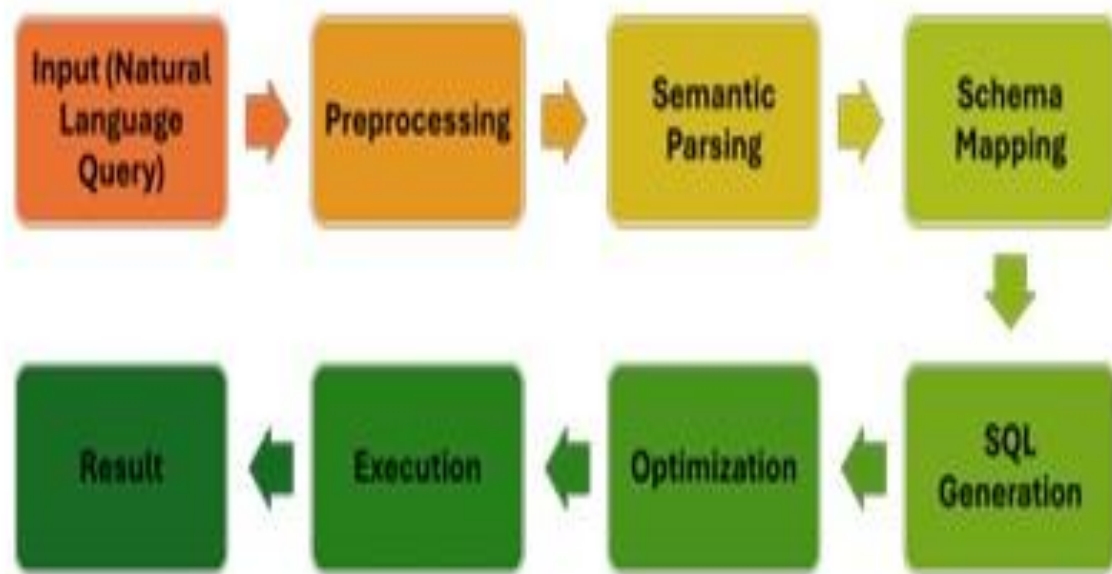


Fig.:2 Text-to-SQL Process Overview

1) Natural Language Understanding (NLU): This involves parsing and interpreting the user's query to comprehend its intent and semantics. Techniques such as tokenization, part-of-speech tagging, and syntactic parsing are employed to analyze the structure and meaning of the input.

2) Schema Linking: This process connects elements of the natural language query to the corresponding components in the database schema, such as tables and columns. Effective schema linking is crucial for accurately mapping user intents to database structures.

3) Semantic Parsing: This step involves converting the natural language query into an intermediate logical form that represents its meaning. Semantic parsing serves as a bridge between the user's intent and the formal SQL query.

4) SQL Generation: The final component translates the intermediate logical form into a syntactically correct and executable SQL statement. This requires understanding SQL syntax and ensuring that the generated query aligns with the database schema.

Advancements in artificial intelligence, particularly in deep learning and natural language processing, have significantly enhanced the performance of text-to-SQL systems. For instance, the integration of large pre-trained language models has improved the systems' ability to understand complex queries and generate accurate SQL statements [4]. Despite these advancements, challenges remain, especially in handling complex and cross-domain queries. Ongoing research focuses on improving the robustness and adaptability of text-to-SQL systems to address these challenges [2]

CHAPTER – 2

LITERATURE SURVEY

In [1], the authors claim that deep learning neural models constitute a challenge to classic generative phrase structure theories of natural language, as they may be applied to software code. They contend that since the same reasoning hold true for programming languages, attaining great performance on random NLP tasks has little bearing on theories of natural language and generative grammar. They suggest a more precise term, “corpus model,” for these models and contend that current neural models fall short in representing pertinent linguistic events. They contend that although artificial neural networks have shown promise in some scenarios, they have not fundamentally changed our understanding of how people think. In [2], in this study, an automated technique for producing user-friendly images— prompt evolution—is presented. It yields notable gains in image quality by fusing the diversity of evolutionary algorithms with the innate capabilities of generative models. Multi X formulations, single-objective settings, and multi-objective settings can all be used with this technique. It also imposes variation and selection pressure on concealed states, guidance, and outputs, providing a wealth of research opportunities in this nascent discipline. In [3], the study underscores the need of AI literacy in enabling cognitive engagement with generative AI, which unleashes creative possibilities. It focuses on how young students engage with AI and how that engages them in critical thinking, creativity, and problem-solving. The study discovered that high school pupils in Northern Thailand had favourable opinions about AI and regarded it to be highly useful. Important features include aiding skill development, encouraging collaboration, encouraging accessibility, encouraging iteration, and stimulating creative thinking. The report highlights how important it is to incorporate AI instruction into curricula. In [4], in order to help the neural network concentrate more on the relevant portions of the input and less on the irrelevant parts during prediction tasks, the attention mechanism gives each input item a weight. Recent years have seen a major increase in interest in this mechanism, which has prompted more study and applications across a range of industries.

In [5], this study presents a new approach to semantic similarity calculation based on multiple influencing elements, which leads to a more accurate result than the original classical algorithm and enhances accuracy and alignment with experts' expertise. In [6], over the past 60 years, natural language processing, or NLP, has advanced dramatically. For example, gadgets are now able to process voice instructions, and Google's machine 26 translation service can translate between numerous languages. This illustrates the capabilities of learning systems and neural networks. Machine-learning frameworks have advanced at a never-before-seen rate, setting records across a range of industries. In natural language challenges, deep neural architecture models perform more efficiently and introduced the idea of "imperfection." Nonetheless, the majority of research ignores speakers of other languages by using English as the input or output language.

Future research ought to take into account utilizing a larger variety of languages and forecast that deep learning models would advance and eventually become the standard in computational linguistics. In [7], In terms of computation and architecture, convolutional neural networks (CNN) have significantly advanced natural language processing (NLP).

Word embeddings are introduced by the Word2vec approach, which enhances NLP performance. Continuous Bag-of-Words (CBOW) and Skip-Gram are two famous algorithms used for learning dense representations of words in a lower-dimensional vector space. These methods are extensively employed in many different NLP applications. In [8], a sophisticated and crucial component of artificial intelligence, natural language processing (NLP) enables chatbots to comprehend human input, convert it into a format that can be processed, and reply to it. To comprehend its complexity, one must have a great commitment and determination. An essential component of contemporary AI systems, interview performance evaluation is anticipated to be automatically assessed as AI advances. In [9], the algorithms for supervised, unsupervised, semi supervised, and reinforcement learning were examined in this paper along with other machine learning techniques and approaches. In the future, it hopes to create a model based on these methodologies, having contributed to their classification

CHAPTER – 3

ARCHITECTURE

The system architecture of the project seamlessly intertwines Natural Language Processing (NLP) and Deep learning, incorporating sophisticated machine learning and advanced linguistic analysis. The primary modules are designed to synergize their functionalities, fostering a cohesive system that transforms raw natural language queries into precise SQL expressions.

The Natural Language Processing (NLP) Module serves as the foundational layer, tasked with processing and analyzing raw natural language queries. This involves intricate processes such as tokenization, which dissects sentences into individual tokens; part-of-speech tagging, identifying grammatical parts of speech for each token; Named Entity Recognition (NER), detecting entities like table names, attributes, and conditions; and Lemmatization, reducing words to their base or root form.

This comprehensive NLP layer forms the basis for the system's understanding of user input. Moving forward, the Intent Classification and Entity Recognition Module complements the NLP layer by determining the user's intent and recognizing pertinent entities. Machine Learning Classifiers, drawing insights from historical data, classify user intent, while Neural Network-based models enhance the system's ability to recognize entities within the processed natural language. This module ensures a nuanced understanding of user queries, refining the context for subsequent stages.

The Generative AI Model, employing a Seq2Seq architecture, is a pivotal component designed to translate processed natural language queries into SQL expressions accurately. The Sequence-to-Sequence (Seq2Seq) Model, functioning as an encoder-decoder architecture, is augmented by an Attention Mechanism, allowing the model to focus adeptly on relevant segments of the input during the translation process. This integration of advanced AI mechanisms bridges the gap between natural language comprehension and SQL query generation.

The Algorithmic Explanation Generation Module complements the system by providing transparent insights into the process of query generation. Rule-Based Algorithms construct explanations based on predefined rules, ensuring users can comprehend the rationale behind the system's decisions. There is potential integration of Natural Language Generation Models, introducing coherent text generation for a more user-friendly explanation interface.

The Cross-Domain Adaptability Module further extends the system's versatility by handling queries from various domains. This module incorporates fine-tuned Domain Specific Language Models, enabling the system to understand and generate queries specific to different domains. Dynamic Training Data Selection algorithms enhance adaptability by selectively incorporating domain-specific training data, ensuring the system's flexibility across diverse contexts.

In essence, the overall system architecture seamlessly integrates these modules, creating a sophisticated framework where NLP serves as the initial processing layer, followed by intent classification, SQL query generation, and algorithmic explanation. This cohesive design ensures a comprehensive and user-centric approach to converting natural language queries into SQL expressions, laying the foundation for an accessible and intelligent database interaction system.

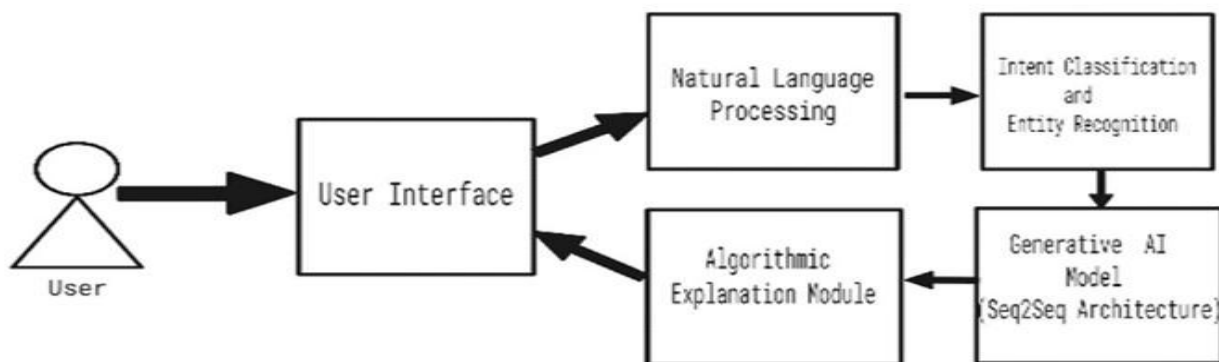


Fig:3 Architecture diagram

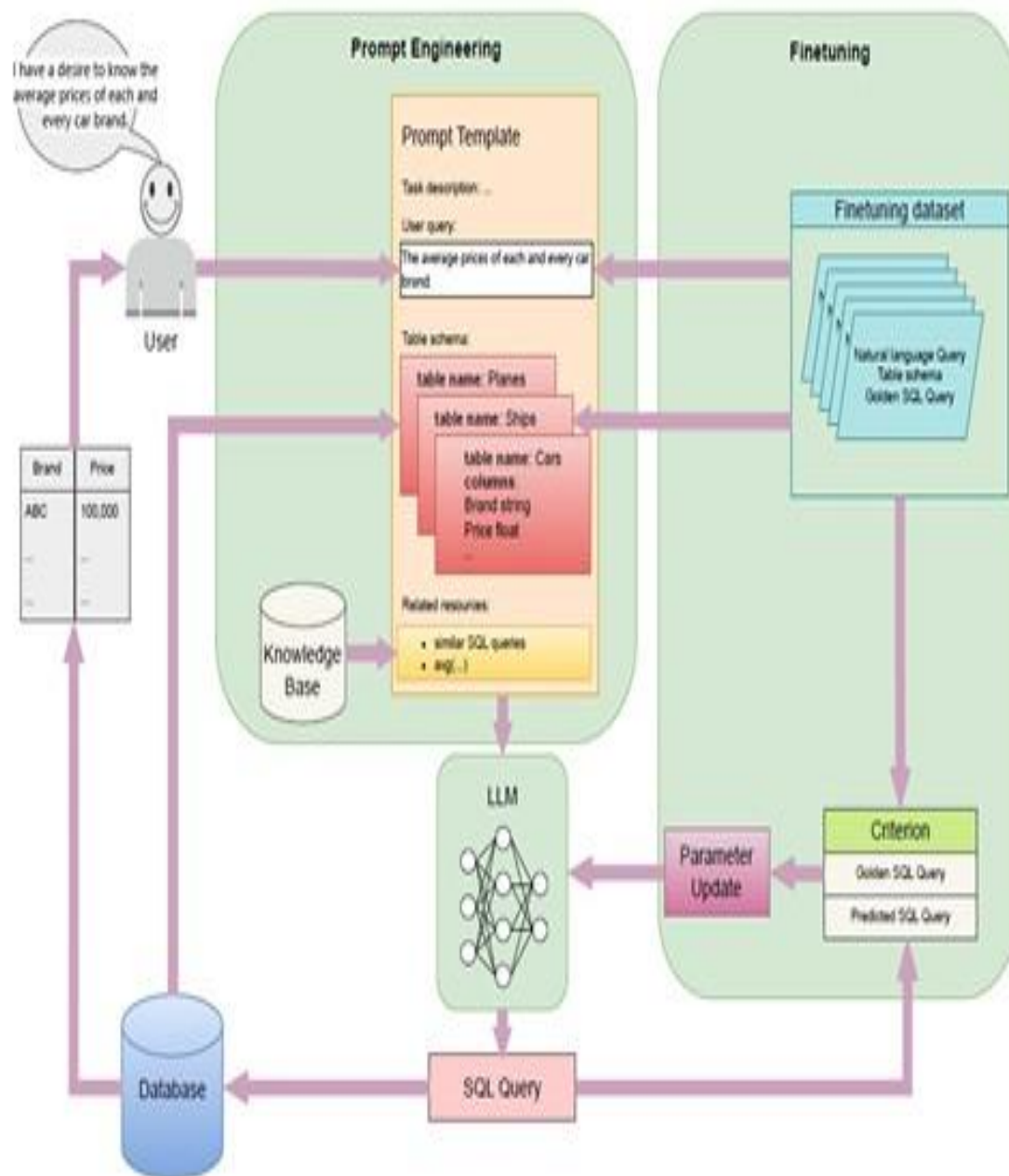


Fig:4 LLM Framework for text to SQL

CHAPTER – 4

PROPOSED SYSTEM

The project addresses a critical need in the contemporary digital landscape, where databases serve as repositories for an ever-increasing volume of information. Traditional methods of interacting with databases, predominantly reliant on SQL proficiency, pose challenges for non-technical and non-proficient users. The proposed solution aims to democratize database access, providing an inclusive platform for individuals across diverse domains to effortlessly formulate complex queries using natural language.

The success of this solution hinges on a meticulously designed set of core mechanisms, each serving a distinct role to advance the overarching objectives of the project. Comprehending and extracting meaningful information from raw natural language queries is achieved through the implementation of advanced Large Language Models (LLM) and Transformers. These algorithms, encompassing tokenization, part-of-speech tagging, named entity recognition (NER), and lemmatization, collectively form a sophisticated framework for processing and analyzing user input.

The objective is clear: enhance the system's ability to decipher the intricacies of human language, paving the way for more accurate and context-aware query interpretation. Moving beyond raw input understanding, the project incorporates Intent Classification and Entity Recognition mechanisms.

Through the implementation of machine learning classifiers and neural network-based models, the system endeavors to unravel the user's intent and pinpoint entities crucial for query construction. This layer of sophistication ensures that the generated queries not only capture the literal meaning of the input but also encapsulate the user's underlying purpose. The crux of the system lies in the Generative AI Model (Seq2Seq), tasked with translating processed natural language queries into precise SQL queries.

Leveraging the Seq2Seq model, fortified with attention mechanisms, this mechanism attains contextual translation, aligning the generated SQL queries with the nuanced intent of the user. This step represents a pivotal bridge between natural language comprehension and SQL query formulation, marking a significant leap in user-friendly database interaction. Ensuring transparency in the query generation process, the Algorithmic Explanation Generation mechanism steps in. By integrating rule-based algorithms and potentially natural language generation models, the system creates detailed explanations for the generated SQL queries.

This not only aids users in understanding the rationale behind the system's decisions but also contributes to a more user-centric and educational interaction paradigm. In essence, the project orchestrates these core mechanisms seamlessly, weaving together the intricacies of language processing, machine learning, and generative artificial intelligence.

CHAPTER – 5

APPLICATIONS AND USE CASES

Text-to-SQL systems serve as a pivotal tool across diverse industries, enabling natural language interaction with databases to facilitate data retrieval, analysis, and decision making. These systems simplify complex data queries, empowering non-technical users to unlock valuable insights from structured data. Table II summarizes the practical applications, challenges, and benefits of Text-to-SQL systems across key domains. Below, we detail specific applications in health care, education, finance, and business intelligence, showcasing the transformative potential of these systems in addressing industry-specific challenges.

A) Healthcare and Medical Records- Data Management

- **Clinical Decision Support:** Text-to-SQL systems assist healthcare professionals in querying patient records, retrieving relevant medical histories, or aggregating patient data for epidemiological studies.
- **Patient Record Management:** Allows healthcare providers to retrieve complex patient information, such as identifying all patients with certain conditions or recent test results, enabling faster access to critical data.
- **Medical Research:** Facilitates large-scale data retrieval from medical databases for research on disease patterns, drug efficacy, or population health studies.

B) Educational Tool

- **Adaptive Learning Systems:** Text-to-SQL can be integrated into educational platforms to create adaptive learning tools that personalize content based on student data stored in databases.
- **Academic Analytics:** Enables querying of student performance data, helping in curriculum assessment, identifying at-risk students, and evaluating the impact of educational interventions.

- **Student Self-Service:** Allows students to query databases for information on course requirements, academic records, or library resources without needing technical expertise.

C) Finance and Banking

- **Financial Reporting and Analytics:** Enables financial analysts and executives to interact with databases for generating real-time financial reports, trend analyses, and risk assessments.
- **Customer Service and Query Resolution:** Empowers customer service agents to query customer data efficiently, providing personalized and accurate information in real-time.

D) Business Intelligence and Analytics

- **Market Analysis and Trend Detection:** Text-to-SQL streamlines data retrieval for analyzing customer behavior, tracking sales performance, and identifying emerging market trends
- **Inventory and Supply Chain Management:** Assists businesses in monitoring inventory levels, supply chain metrics, and order statuses to improve operational efficiency.
- **Employee Productivity and HR Analytics:** Enables querying workforce management data, allowing companies to monitor employee productivity, analyze turnover rates, or assess training needs.
- **LinkedIn's SQL Bot- Data Democratization:** A multi agent system built on LangChain and LangGraph, SQL Bot empowers LinkedIn employees to transform natural language questions into SQL queries, automating workflows, reducing dependencies on data teams, and accelerating decision-making across diverse business functions.

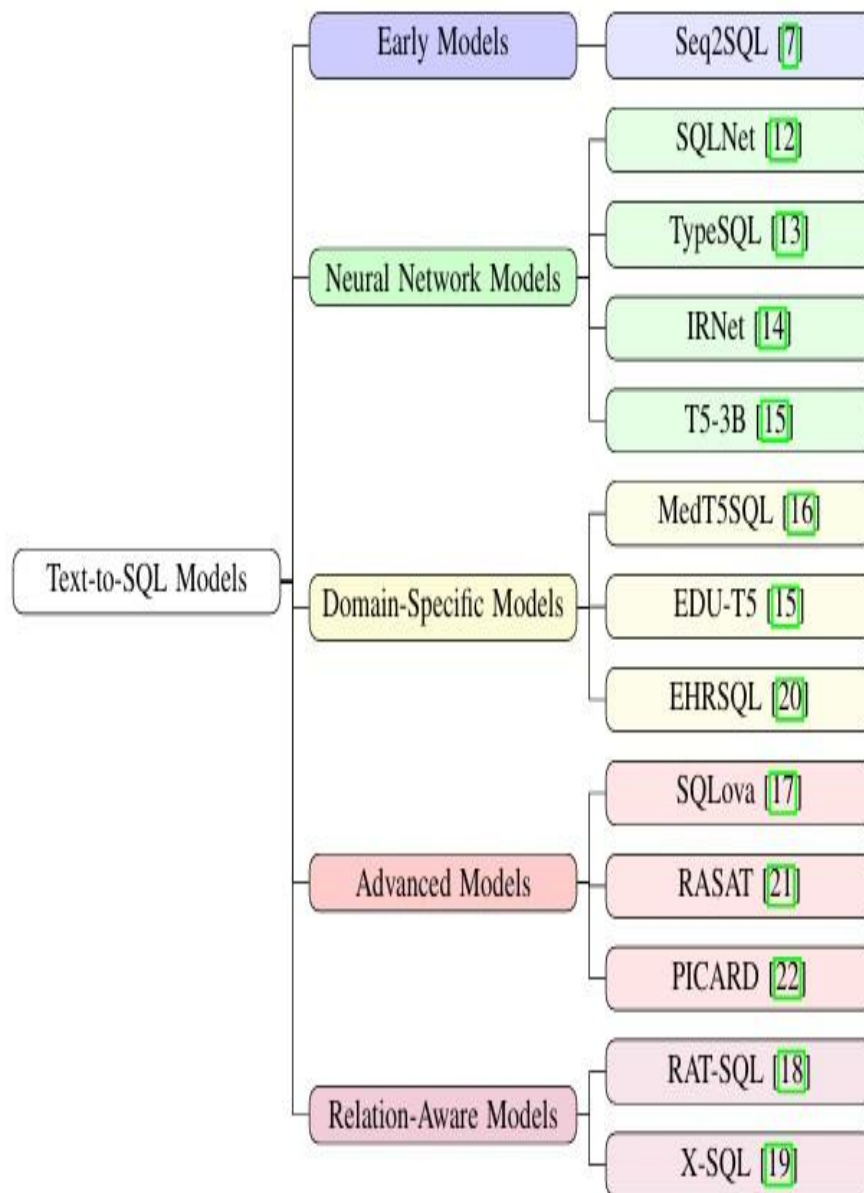
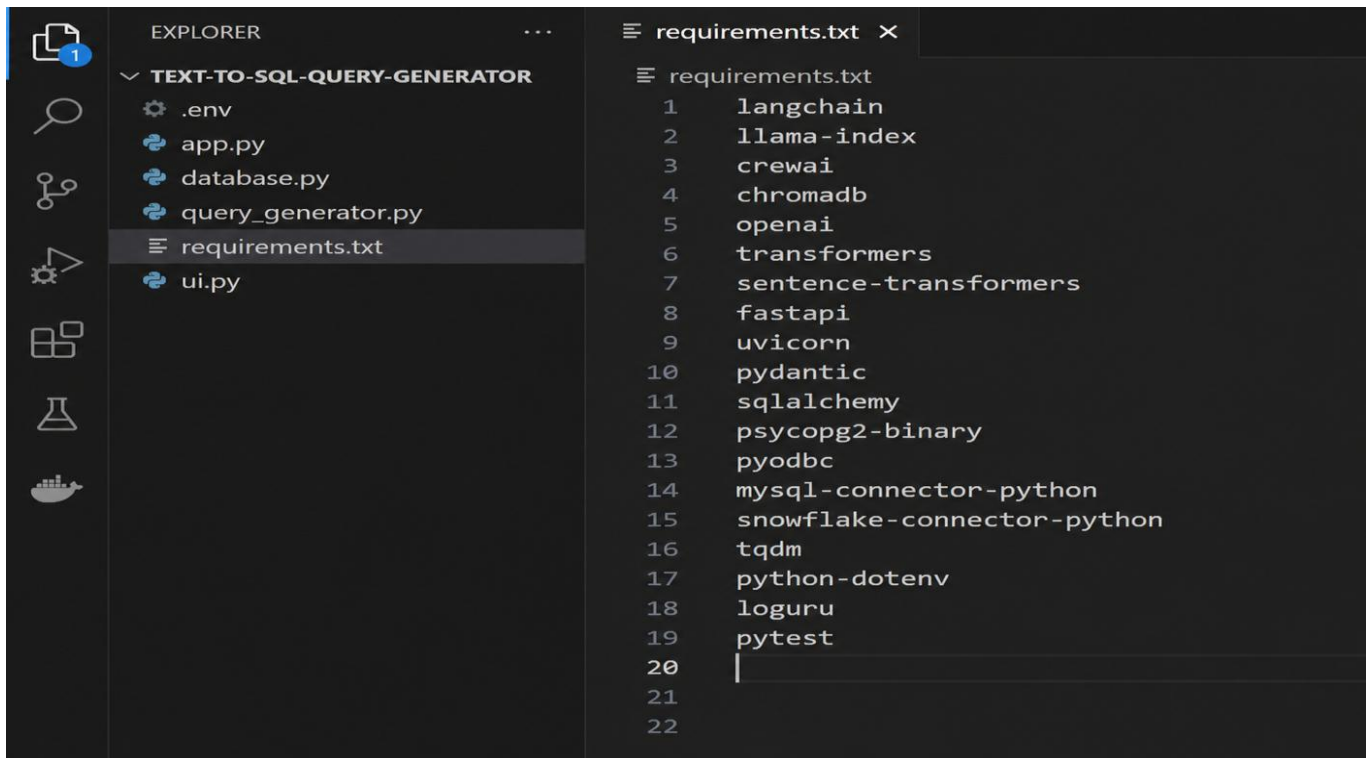


Fig: 5 Hierarchical Tree of Text-to-SQL Model

CHAPTER - 6

CODE IMPLEMENTATION

6.1 PACKAGE REQUIREMENTS



6.2 . ENV

```
OPENAI_API_KEY=your_openai_api_key_here
```


CHAPTER – 7

CHALLENGES AND FUTURE DIRECTIONS

Despite significant advancements, several challenges remain in the development of text-to-SQL systems. Future research aims to address these challenges through the following directions.

A) Industry-Specific Challenges and Solutions

Healthcare:

Handling highly sensitive patient data requires robust privacy-preserving techniques in text-to-SQL systems.

Queries often require contextual medical knowledge not present in the database schema.

Finance:

The prevalence of NoSQL databases adds complexity to adapting text-to-SQL systems

Ambiguities in financial terminologies, such as risk related queries, require domain-specific language models.

Education:

Diverse grading structures and course formats lead to variability in database schema, complicating schema linking.

Lack of standardized datasets reflecting academic systems globally limits model generalization.

B) Generalization Across Domains

Challenge:

Many text-to-SQL models struggle with do main adaptation, particularly when faced with unfamiliar database schemas or industries

.

Future Direction:

Developing universal or multi-domain models that can interpret and generate accurate SQL for varied domains without additional retraining or extensive customization.

C) Handling Ambiguity in Natural Language**Challenge:**

Natural language queries can be ambiguous or lack specific details required for accurate SQL generation (e.g., missing parameters, vague terms).

Future Direction:

Implementing advanced disambiguation techniques or interactive clarification processes, where the model seeks clarification for ambiguous inputs to ensure query accuracy.

C) Incorporating External Knowledge**Challenge:**

Domain-specific queries often require contextual knowledge not present in the database schema, such as industry-specific terms or common data patterns.

Future Direction:

Integrating external knowledge bases or ontologies to enhance the model's contextual understanding and allow it to interpret queries with a broader scope.

E) Optimizing SQL Efficiency**Challenge:**

While accuracy is essential, the efficiency of SQL queries is also important, especially for large databases where complex queries can be resource intensive.

Future Direction: Researching optimization techniques for SQL generation, such as index-aware querying or using machine learning to predict and avoid performance bottlenecks in query structures

F) Human-in-the-Loop Systems

Challenge

Fully automated systems may not meet the nuanced requirements of real-world applications.

Future Direction:

Developing interactive text-to-SQL systems where users can validate or edit SQL outputs, providing an extra layer of accuracy and flexibility for complex queries.

G) Improved Interpretability and Debugging

Challenge:

Debugging and interpreting complex SQL statements generated by AI models can be challenging, particularly in mission-critical applications

Future Direction:

Focusing on interpretability in model design, enabling end-users to understand how the generated SQL queries relate to the original natural language inputs, potentially through visual representations or explanations.

CHAPTER – 8

RESULTS

Upon completion of the project “Conversion of Natural Language Text to SQL Queries using Generative AI”, the system is remarkably adept at enabling user-friendly database interaction. Users input queries in natural language, and the system swiftly crafts precise SQL queries based on the schema provided by developers. Moreover, by deconstructing the methodical process of query construction into algorithmic descriptions, the system offers users priceless insights. The dual-layered output not only secures precise translation from natural language to SQL but also amplifies user comprehension and confidence in the system’s capabilities. By seamlessly bridging the semantic gap between user-conveyed intentions and SQL syntax intricacies, the project successfully democratizes database interaction, empowering individuals across diverse domains to effortlessly extract insights and leverage the power of relational databases with unprecedented ease and efficiency. Table 1 illustrates the superiority of the proposed system over existing systems in terms of accessibility, approach, algorithmic description, and adaptability to domains. It demonstrates how the proposed system outperforms its counterparts by offering a more user friendly interface, a different approach, comprehensive algorithmic descriptions, and the ability to seamlessly adapt to various domains with ease.

Aspect	This project	Existing project
Accessibility	Enhances accessibility by offering a natural language interface,empowering users with limited proficiency to interact with databases	Accessibility features may vary ,with some systems lacking intuitive interfaces or support for non-technical users.
Approach	Utilizes Generative AI for generating SQL queries when natural language input is given	Various approaches,including rule- statistical , and machiene learning methods.
Algorithmic Description	Provides algorithmic breakdowns for transparency and understanding	May lack detailed explanations or provide non – specific and limited insights into the query generation.
Adaptability to Domains	Demonstrates adaptability across various domains and industries.	May have limited adaptability to specific domains , requiring extensive customization.

Table: 1 Differences between Proposed System and Existing System

SAMPLE OUTPUTS



Fig : 6 User interface of working



Fig : 7 Schema Input

SQL Generator

User's Prompt

What is the total amount spent by customers with a postal code starting with "7"?

Submit Prompt

Query

```
SELECT SUM(p.price)
FROM customer c
JOIN product p ON c.id = p.id
WHERE c.postalCode LIKE '7%';
```

Algorithm

Start

1. Start by selecting the sum of product prices from the product table.
2. Join the product table with the customer table on the customer id.
3. Filter for customers with postal codes starting with '7'.
4. Use the LIKE operator to specify the pattern to match.
5. Perform the sum function on the product prices. End

Fig : 8 SQL Query Generation

CONCLUSION

In conclusion, the project “Conversion of Natural Language Text to SQL Queries using Generative AI” represents a significant advancement in the field of database interaction and natural language processing (NLP). Through the integration of advanced generative AI techniques and deep learning models, the system successfully bridges the semantic gap between user-conveyed intentions and the intricacies of structured query language (SQL). The project achieves its core objective of democratizing database interaction by empowering individuals with limited SQL proficiency to seamlessly formulate queries in natural language, thereby enhancing accessibility and usability across diverse domains and industries. The construction of end-to-end applications using Generative AI, exemplified by the “Conversion of Natural Language to SQL Queries” project, holds immense promise in revolutionizing the landscape of human-computer interaction. The results obtained under score the significance and potential impact of such applications. The precision achieved in natural language processing, coupled with accurate intent classification and entity recognition, demonstrates the system’s adeptness in understanding and interpreting user queries. The performance of the Generative AI Model, particularly its ability to translate natural language queries into SQL expressions, marks a significant advancement in making complex database interactions accessible to a broader audience. Additionally, the provision of detailed algorithmic breakdowns enhances transparency and fosters user understanding of the query generation process, instilling confidence in the system’s capabilities. The results obtained from this project lay the groundwork for future advancements in end to-end applications utilizing Generative AI. As we look ahead, the integration of such technologies opens doors to new possibilities, empowering users with sophisticated tools that bridge the gap between human language and machine understanding. This research paper serves as a testament to the potential of Generative AI in constructing intelligent, user centric applications, setting the stage for a transformative era in human-computer interaction.

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